

EDUCATION

University of Warwick — MDatSci, Statistics & Machine Learning Sept 2023 – Present

- First Class average of 75.5% across the first three years
- Awarded the Department of Computer Science's Best Data Science Third-Year Prize, for the brglm2 open-source contribution (CS350 Data Science Project, 89/100)

Denstone College — A-Levels (A*AAA) Sept 2021 – June 2023

- Maths, Further Maths, Physics, Computer Science

Denstone College — GCSEs (11 A*) Sept 2018 – June 2021

EXPERIENCE

BNY — Engineering Developer, Foundational Frameworks Team

- Building an agent-driven diagnostic tool spanning LLM prompt engineering through to MCP server tooling for complex reasoning, on the Java Spring AI framework
- Own the stack end-to-end: designed and built the API endpoints and deployment pipeline; demoed the system to 100+ engineers and stakeholders

WorldQuant — BRAIN Consultant

- Developing and backtesting alpha signals for robust market strategies, evaluating performance through Sharpe and fitness ratios. Focused on uncovering insights that drive smarter, data-backed trading decisions

Warwick Data Science Society — President

- Lead a 15-member executive team, ensuring events run smoothly and initiatives have impact. Maintain close ties with partner departments (Statistics, Mathematics) and sponsors including Jane Street and WorldQuant

Warwick Data Science Society — Outreach Head

- Secured partnerships with industry sponsors and coordinated speakers and events, connecting 80+ students with professionals

Warwick Mixed Hockey Club — Training Secretary

- Provided fitness and training sessions three times a week for 100+ members, tailored across three ability levels

Warwick Mixed Hockey Club — Fresher Representative

- Elected as one of two Fresher Representatives; communicated ideas and opinions effectively with the executive committee

PROJECTS

Diffusion Models for Implied Volatility Surfaces MDatSci Dissertation — in progress

- Supervised by Prof. Paul Jenkins, Department of Statistics, University of Warwick
- Researching score-based (DDPM) generation of arbitrage-free implied volatility surfaces, building on Jin & Agarwal's approach and benchmarking hedging performance against Black-Scholes, historical volatility, and VolGAN
- Investigating whether exact conditional simulation via forward-backward SDE bridging can replace the standard approximate learned-conditional-drift approach used in DDPMs

brglm2 — Open-Source Contribution to Bias-Reduction Methods for GLMs 3rd-Year Data Science Project — 89/100

- Implemented a trust-region estimation algorithm as an alternative fitting method for brglm2, an actively maintained CRAN package for bias reduction in generalised linear models
- Contribution is publicly available and pending the package maintainer's review ahead of an upstream merge into CRAN
- Awarded the Department of Computer Science's Best Data Science Third-Year Prize for this work

Databases & SQL Coursework

85%

- Implemented database applications in PostgreSQL via JDBC in Java. Developed and optimised complex SQL queries with views, stored procedures, CTEs, and transactions for real-world use cases such as ticketing, scheduling, and cancellation workflows

Python for Data Science Course

Warwick Data Science Society

- Designed and delivered a Python for Data Science course covering data manipulation, exploratory data analysis, and predictive modelling

AWARDS & CERTIFICATIONS

- Department of Computer Science — Best Data Science Third-Year Prize (brglm2 project)
- WorldQuant — IQC National Finalist: one of 8 teams to reach the national final (ranked 39th globally out of 37,000+ entrants); presented quantitative models to a judging panel and fielded a live Q&A
- WBSS — Quantitative Investment Group Academy: relative valuation using financial ratios, statements, and multiples

SKILLS

Languages & Tools: Python, R, SQL, Java, C++, C#, LaTeX

Applied AI / Agentic Tooling: LLM prompt engineering, MCP server tooling, Java Spring AI

INTERESTS

Investing, Research, Programming, Field Hockey, F1

EXTRA-CURRICULAR

- Duke of Edinburgh: Bronze, Silver, and Gold (in progress)
- 7 years of voluntary work for a local Cub Scouts programme